

Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub

What the threshold model predicts—and what the design cannot identify

Gabriel Saco

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Universidad del Pacífico

Quispe & Xu (2026) · arXiv:2605.25438

github.com/gsaco/ai-03-quispe/tree/branch1

The paper and the developer's problem

Question and economic mechanism

Question. Does an agent expand the set of programming languages in which a developer can ship code? This is a **production frontier**, not a claim that the developer learns each language.

Mechanism. Conversational AI augments existing language skill; agentic AI adds delegated execution under human specification and verification.

Choice problem for developer i , language k , month t

Pre-agent menu $\mathcal{M}_1 = \{S, C\}$; post-agent menu $\mathcal{M}_2 = \{S, C, D\}$:

$$m_{ikt}^* \in \arg \max_{m \in \mathcal{M}_g} V_{ikt}^m, \quad Z_{ikt}^g = \mathbb{I}(\max_m V_{ikt}^m \geq 0).$$

Monthly breadth is $N_{it}^g = \sum_k Z_{ikt}^g$.

S : own execution.

C : suggestion gain $\gamma s - r_C$.

D : agent execution $\lambda az(A)$, net of verification $\kappa(a, s)$, compute r_D , and residual risk.

Main result: a conditional activation band

Proposition 2 for an unfamiliar language

If conversational assistance has no value without a skill foothold, $\gamma \underline{s} - r_C \leq 0$, then $T^1 = T^S$. If delegation lowers the threshold, $B \equiv T^S - T^D > 0$, then

$$Z^2 - Z^1 = \mathbb{I}(T^D \leq \omega < T^S), \quad \Pr(Z^2 - Z^1 = 1) = F(T^S) - F(T^D).$$

Conditions that do the work

Theory

- ▶ no conversational foothold in k ;
- ▶ verification, compute, and residual-risk costs leave $B > 0$;
- ▶ opportunity mass lies inside $[T^D, T^S)$;
- ▶ the developer can specify and verify.

Empirical interpretation

- ▶ public, sustained adopters; “new” since Jan. 2024;
- ▶ not-yet adopters are valid counterfactuals;
- ▶ no time-varying project shock jointly moves adoption and languages.

Observed at adoption: +2.53 active languages and +1.19 newly used languages; these are event-time associations, not identified causal effects.

What I did: one modelling choice, two direct answers

Quispe-Xu: production expands

Language skill s_{ik} is a **fixed state** over the choice horizon. Claude adds delegation to the menu and lowers the entry threshold:

$$\{S, C\} \longrightarrow \{S, C, D\}, \quad T^D < T^S.$$

Result: more languages become feasible. This is explicitly a **production frontier**, not learning.

Aouad-Lykouris-Zhong: skills can erode

Skill, effort, and AI are substitutable inputs:

$$x_t = s_t + e_t + a_t, \quad a_t \uparrow \Rightarrow e_t^* \downarrow.$$

They then make future skill depend on current effort. With myopic users and a positive sensitivity gap, lower effort shifts the steady-state skill distribution downward and productivity can fall.

The decisive choice is the skill law of motion: fixed skill gives static frontier expansion; effort-dependent skill turns substitution into deskilling.

Q1 • Does the threshold require an agent? No.

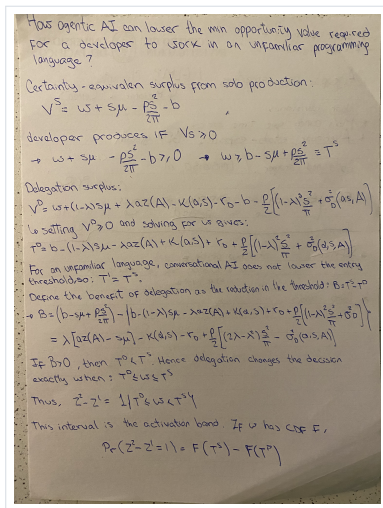
Any positive generic gain that lowers T^1 produces the same activation band. Agent specificity comes from the mode assumptions, not from the threshold algebra.

Q2 • Does the design identify expansion? No.

A new-language project can trigger adoption and breadth together. Flat pre-trends and outcome exclusions do not make adoption timing exogenous.

Aouad, Lykouris & Zhong (2026), arXiv:2605.11350. With exogenous skill transitions, their productivity paradox disappears.

Hand and Lean check: where I did not believe the AI



My handwritten threshold and selection derivation.

Lean result 1 • threshold non-uniqueness

For every $T^D < T^1$, Lean constructs a generic gain $g > 0$ such that $T^1 - g = T^D$. The lower threshold and its activation band are therefore not uniquely agentic.

Lean result 2 • zero-effect selection

Lean also verifies the sign implication

$$\tau = 0, \beta > 0, P_A > P_C \implies \widehat{ATT}(0) > 0.$$

A project shock can create adoption and diversification together while all earlier differences remain flat.

Verdict: coherent mechanism, unidentified cause

Quispe-Xu can explain production expansion even while the Aouad-Lykouris-Zhong feedback predicts deskilling. The data measure the first outcome, not the missing skill stock, and voluntary adoption still confounds delegation with project selection.